

Turbo Equalization for Single-Carrier Underwater Acoustic Communications

Yahong Rosa Zheng, Jingxian Wu, and Chengshan Xiao

ABSTRACT

Recent research in underwater acoustic communications has taken advantage of MIMO technologies to achieve reliable communication with 10–100 times increase of data rate in comparison to traditional systems. The powerful turbo equalization and FEC coding techniques enable both single-carrier modulation and OFDM systems to combat triply selective UWA channels. This article reviews the time-domain and frequency-domain turbo equalizer schemes for MIMO SCM systems. Low-complexity techniques are presented with both turbo linear equalizers and turbo soft decision feedback equalizers in both the time and frequency domains. Although results are shown specifically for UWA channels, these turbo equalizer techniques are also suitable for terrestrial RF communication systems.

INTRODUCTION

Underwater acoustic (UWA) communication, when integrated with underwater robots, unmanned vehicles, or wireless sensor networks, plays a critical role in ocean exploration, underwater infrastructure monitoring, disaster prevention and rescue, and military surveillance. Acoustic wave is the most energy-effective means to achieve medium (1–10 km) and long (10–1000 km) range communications. However, UWA channels exhibit significant technical challenges for achieving high throughput because of their limited bandwidth, slow wave propagation, severe noise and interference, and time-varying dispersion. In contrast to radio frequency communications, common horizontal UWA channels have about 100 kHz bandwidth for 1 km range or 40 kHz for 10 km range due to frequency-dependent attenuation; while an RF wireless channel usually occupies 1–20 MHz bandwidth. Over the last decade, coherent UWA communication has achieved data rates on the order of 10 kb/s [1], and recent application of multiple-input multiple-output (MIMO) technologies in UWA has boosted the rate in excess of 100 kb/s for medium-range communications.

MIMO UWA channels experience severe

triply-selective fading. Extremely long delay spread is often on the order of 10 ms, which results in inter-symbol interference (ISI) of 100 taps at 10 ksym/s). The maximum Doppler spread is on the order of 1–10 Hz due to the motion of water as well as transceiver platforms. Severe Doppler spread causes fast time-variation in channel impulse responses (CIRs). Inversely proportional to the maximal Doppler frequency, the coherence time of the UWA fading channels can be as short as 100 ms or 1000 symbols at 10 ksym/s. Simultaneous transmission by multiple elements at the transmitter can boost data rate on one hand, but also causes co-channel interference (CCI) on the other hand. Therefore, powerful MIMO receiver techniques are required to enable reliable reception of high data rate communication.

Turbo equalization is one of the enabling technologies for wideband RF communication and has recently been applied to UWA transceivers [1]. In contrast to conventional equalization and detection schemes, which make hard decisions in one iteration, turbo equalizers exchange the extrinsic information of soft symbols/bits with the soft decision decoders iteratively. Since forward error correction (FEC) coding embeds correlation among the coded bits, and the fading channel introduces correlation among received symbols, turbo equalizers can glean more information about each transmitted bit at each iteration by exploiting the correlation property via adaptive filters.

Due to severe Doppler spread and multipath delay spread in UWA channels, turbo equalizers are often required to achieve satisfactory performance in both single-carrier modulation (SCM) and (multi-carrier) orthogonal frequency-division multiplexing (OFDM) systems [2, 3]. In an SCM system, a symbol is represented by a baseband pulse that occupies the entire bandwidth of the channel and is modulated to a single pass-band carrier; in an OFDM system, the channel bandwidth is divided into hundreds of frequency bins, and a baseband symbol is modulated onto a narrowband carrier. The block of modulated multi-carriers is then converted to time-domain via inverse fast Fourier transform (IFFT), and a transmitted OFDM symbol occupies the entire time duration of the channel.

Yahong Rosa Zheng and
Chengshan Xiao are with
the Missouri University of
Science and Technology.

Jingxian Wu is with the
University of Arkansas,
Fayetteville.

Accurate channel estimation is critical to the performance of the CE-based Turbo equalizers and is achieved by two modes: the training mode uses the pilot block to estimate the channel iteratively, and the decision-directed mode uses the decisions of the previous and current sub-blocks to estimate and track the MIMO channel.

This article focuses on turbo equalization techniques for SCM systems that utilize continuous transmission rather than block transmission [4–7]. Since UWA channels are quite long in comparison to channel coherence time, continuous transmission eliminates zero padding (ZP) or cyclic prefix (CP) required in block transmission. Therefore, continuous transmission achieves better data efficiency than block transmission. In the receiver, the continuous signal stream may be divided into overlapped sub-blocks for equalization in either the time or frequency domain.

Both time-domain and frequency-domain turbo equalizers are divided into two categories: linear equalizer (LE) and decision feedback equalizer (DFE) [8]. Turbo LEs use only feed-forward (FF) filters to filter the received signals, and the filter coefficients are adapted to minimize the mean squared error (MSE) between the equalized symbols and the transmitted ones. Turbo DFEs utilize FF filters to process the received signal and feedback (FB) filters to filter the soft decision symbols of the soft FEC decoders from the previous iteration. The minimum MSE (MMSE) criterion is used for turbo DFE to jointly design the FF and FB filters. Turbo DFE utilizes the soft decisions to reconstruct and remove the residual ISI in the FF filter output, thus achieving better performance than the turbo LE. If designed properly, turbo DFE also reduces catastrophic error propagation often encountered in hard-decision DFE at low signal-to-noise ratio (SNR).

Furthermore, depending on how the adaptation is implemented, turbo equalizers can be divided into channel estimation (CE)-based and direct adaptation (DA)-based ones [9]. The CE-based turbo equalizer explicitly estimates the fading channel coefficients, and then designs the equalizer filters using the channel coefficients and training sequence. The DA-based turbo equalizer adapts the filters directly through the training sequence without estimating the channel coefficients. The CE-based approach requires slightly higher computational complexity but a shorter training sequence and faster convergence than the DA-based approach. In both approaches, the training sequence consists of known pilot symbols and detected symbols from previous sub-blocks.

As DA-based turbo equalizers have been well covered in the literature [1, 9], this article aims to introduce CE-based turbo equalizers that have been applied to MIMO UWA successfully in recent years. For time-domain turbo equalizers, we review the turbo LE [4] and three turbo DFE schemes: the block decision feedback equalizer (BDFE) [5], soft decision feedback equalizer (SDFE) [6], and soft interference cancellation equalizer (SICE) [7]. For frequency-domain turbo equalizers, we first introduce the CP reconstruction technique that enables the low-complexity implementation of FD turbo LE for continuous transmission [7], and then review three FD turbo DFE approaches: frequency domain equalization with time-domain decision feedback (FDE-TDDF), with frequency-domain decision feedback (FDE-FDDF), and time-domain equalizer with FD decision feedback

(TDE-FDDF) [8]. We demonstrate the performance of these schemes by experimental results using the data recorded in the SPACE08 ocean experiment.

In a short article like this, it is difficult to exhaustively cover all research progress in the literature, or cover the technical details in great depths. Interested readers are referred to the references listed here and the references therein. It is our hope that this article provides a study guide for learning and analyzing the complicated MIMO turbo equalization schemes.

SINGLE-CARRIER MIMO SYSTEMS

Assume an $N \times M$ MIMO UWA communication system, as shown in Fig. 1a, where N and M are the numbers of transmit transducers and receive hydrophones, respectively. At the transmitter side, each bitstream $\mathbf{a}_n$ is independently encoded by an FEC encoder. The coded bits $\mathbf{b}_n$ are interleaved and mapped to Q -ary quadrature amplitude modulation (QAM) or phase shift keying (PSK) symbols, and then transmitted after frame construction and carrier modulation. Note that $\mathbf{c}_n$ denotes the interleaved bit vector that maps to symbol s_n , and $\mathbf{s}_k$ denotes the symbol vector at time instant k containing the N transmit symbols.

The received baseband signals are processed iteratively by the turbo MIMO equalizer and decoders. The equalizer outputs the extrinsic log-likelihood ratios (LLRs) of the interleaved bits for each transmit branch $\lambda_e(\mathbf{c}_n)$, for $n = 1, \dots, N$, which becomes the a priori information for the soft-decision (SD) decoder after passing through the de-interleaver. The n th SD decoder outputs the extrinsic information about the coded bits $\lambda_e(\mathbf{b}_n)$, which in turn becomes the a priori LLR of the equalizer after the interleaver. The a priori LLR $\lambda_a(\mathbf{c}_n)$ is used by the turbo equalizer for the next iteration. The iterative process allows the turbo equalizer and the SD decoders to exploit the correlation among the received signals and glean soft information of the transmitted bits back and forth. If the turbo equalizer is designed correctly, the LLRs converge after a few iterations, and the SD decoders make hard decisions $\hat{\mathbf{a}}_n$ on the information bits.

The structure of the transmitted data packet is depicted in Fig. 1b, where the linear frequency modulation (LFM) blocks at the beginning and end of the packet are used for coarse synchronization and Doppler estimation. The m -sequence is used for fine synchronization and initial channel estimation. The data packet is divided into data frames with each frame consisting of a pilot block and a large payload block. In the receiver, the payload block is divided into overlapped sub-blocks for equalization. Accurate channel estimation is critical to the performance of the CE-based turbo equalizers and is achieved in two modes: the training mode uses the pilot block to estimate the channel iteratively, and the decision-directed mode uses the decisions of the previous and current sub-blocks to estimate and track the MIMO channel.

For single-carrier systems, it is unnecessary to insert gaps or cyclic prefixes between the sub-blocks; thus, the data efficiency is higher than

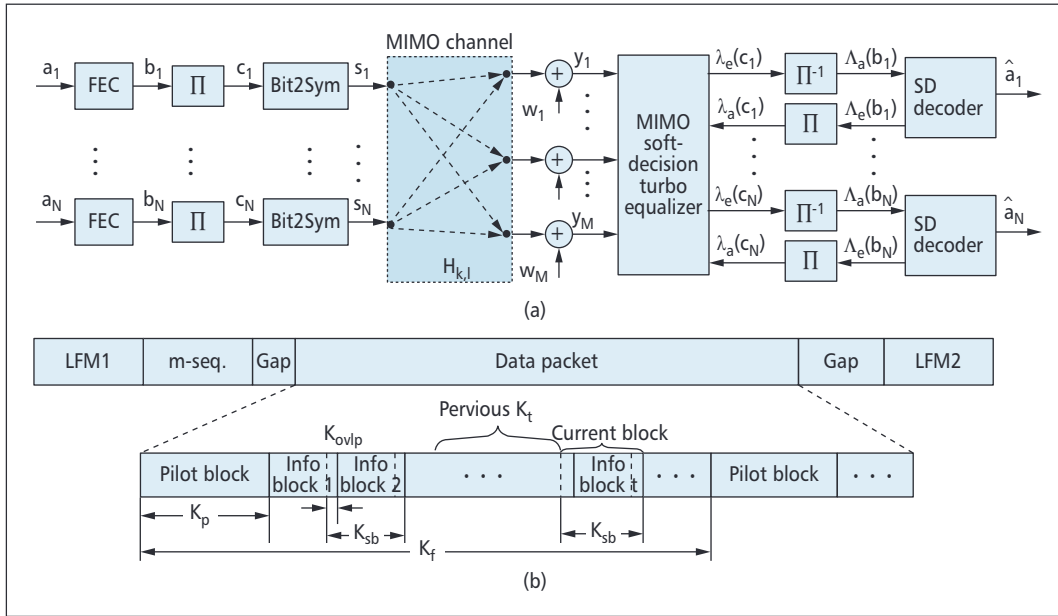


Figure 1. a) Structure of a MIMO communication system with turbo equalizer and maximum *a posteriori* probability (MAP) decoder; b) signaling structure of each transmit branch and subblock format used by the turbo receiver.

The optimal solution often requires the auto-covariance matrix of the equalized/detected soft symbol vector and the cross-covariance of the detected symbol vector and its corresponding transmitted vector. However, the statistics are often difficult to estimate and the receiver has to make reasonable approximations and assumptions.

the block transmission commonly used in OFDM systems. Since no gaps are inserted between sub-blocks, the inter-block interference (IBI) has to be removed from the received signals, and the IBI-free signals corresponding to each sub-block are fed to the turbo equalizer.

The triply-selective MIMO channel at time instance k is modeled as a set of tapped delay lines (TDLs) in which the l th delay tap is an $M \times N$ matrix $\mathbf{H}_{k,l}$. MIMO UWA channels are often noncausal, as shown in the example in Fig. 2. The channel impulse response (CIR) has a main tap that corresponds to the tap of the strongest gain. The main tap divides the overall CIR into causal and anticausal portions. Denoting the lengths of the overall channel, the anticausal CIR, and the causal CIR as L , L_1 , and L_2 , respectively, we have $L = L_1 + L_2 + 1$.

The received baseband signal vector $\mathbf{y}_k$ is the sum of the N transmit branches convolved with the channel and the additive Gaussian noise

$$\mathbf{y}_k = \sum_{l=-L_1}^{L_2} \mathbf{H}_{k,l} \mathbf{s}_{k-l} + \mathbf{w}_k, \quad (1)$$

where $\mathbf{w}_k$ is the vector of M noise components having zero mean and variance σ_w^2 . The long multipath CIRs result in severe ISI at the receiver. The causal CIR causes postcursor ISI, which comes from the past symbols, and the anticausal CIR causes precursor ISI, which is due to the future symbols.

Both time-domain (TD) and frequency-domain (FD) turbo equalizers have been developed for single-carrier UWA MIMO systems. These equalizers are further divided into two types: soft decision linear equalizers (SD-LEs) and soft decision feedback equalizers (SD-DFEs). Based on how the LLRs are calculated in the turbo equalizers, both SD-LEs and SD-DFEs may compute the LLRs using a

sequence estimator that computes the *a posteriori* probabilities based on a sequence of equalized symbols, or a symbol estimator that only uses the current soft symbol to compute the extrinsic LLR of the current symbol. The next two subsections review the TD and FD turbo equalizers that have been successfully applied to UWA.

TIME-DOMAIN TURBO EQUALIZATION

To equalize the MIMO channel in the time domain, the receiver has to first align the current cursor with the main tap CIR, and then design adaptive transversal filters to minimize the MSE between the equalized symbols and the transmitted ground truth. The optimal solution often requires the auto-covariance matrix of the equalized/detected soft symbol vector, and the cross-covariance of the detected symbol vector and its corresponding transmitted vector. However, the statistics are often difficult to estimate, and the receiver has to make reasonable approximations and assumptions. Therefore, it is challenging to ensure convergence and achieve good performance, especially when the receiver has low SNR. Four time-domain turbo equalizers have been developed for UWA communications, as shown in Fig. 3, which can effectively combat the long multipath and severe Doppler channels. This section details each of the TD turbo equalizers, and then compares their complexity and performance.

TD TURBO LE

A time-domain linear equalizer is an FF filter that adapts the filter coefficients to minimize the MSE between the transmitted true symbols and the equalized symbols. Figure 3a depicts a

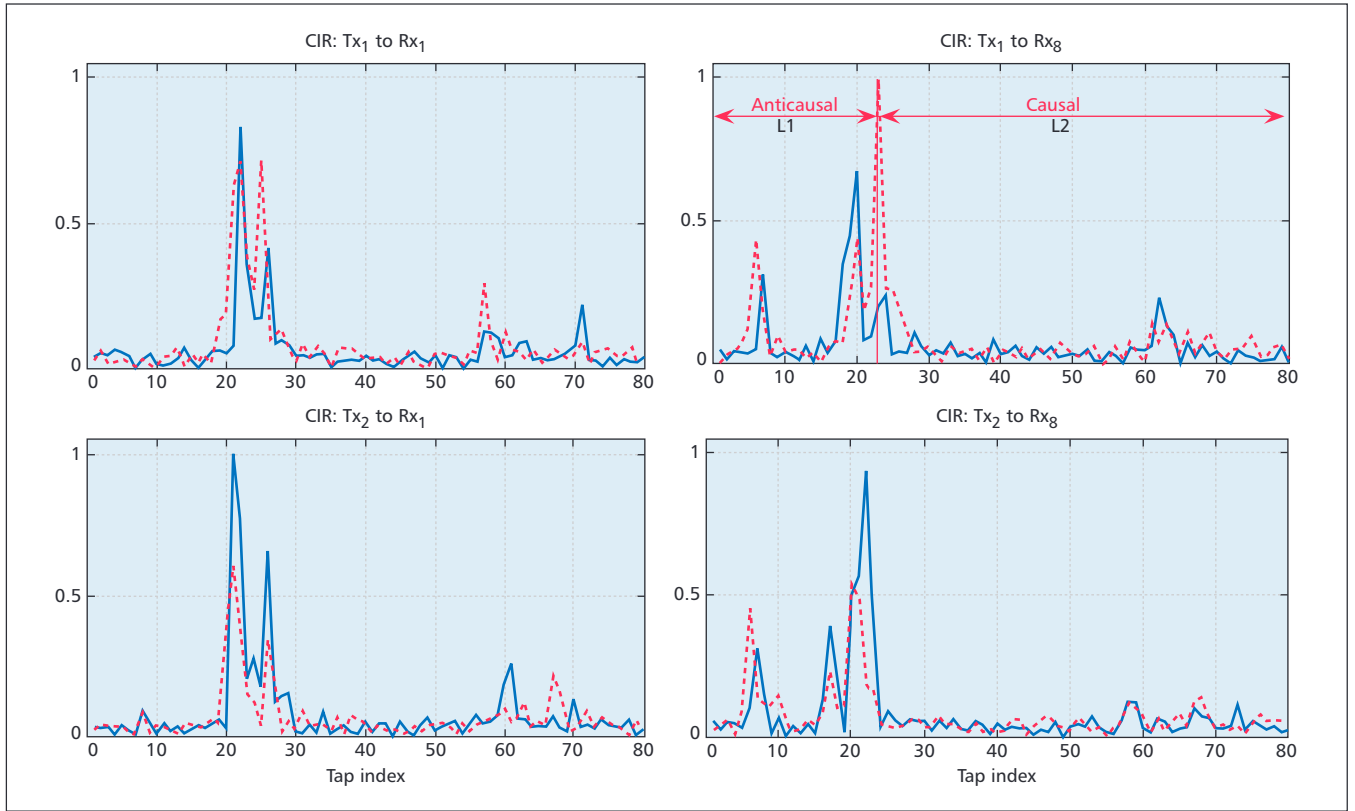


Figure 2. Magnitude of MIMO UWA channel impulse responses at two different time instants, estimated from the data recorded in the GOMEX2008 experiment.

block-based MIMO turbo LE [4] that utilizes hybrid soft interference cancellation. The received signal $\mathbf{y}$ of each sub-block is preprocessed to remove the interference from the previous sub-block. Assume the sub-block length is L_s and the equalizer filter for the n th transmitter branch is $\mathbf{f}_n$, which has $L_f = L_a + L_c + 1$ coefficients, with L_a and L_c being the lengths of the anticausal and causal taps, respectively. The IBI-free signal $\mathbf{y}_k$ at time instant k is a concatenated vector of length ML_f . To equalize the symbol corresponding to the n th transmit branch and the k th time instant, the soft interference from other transmitted branches and time instants is reconstructed by convolving the estimated MIMO channel $\hat{\mathbf{H}}$ with the masked soft symbol vector $\bar{\mathbf{s}}_{n,k}$, the elements of which are the soft symbols previously detected. The precursor (or future) symbols $\bar{\mathbf{s}}$ are calculated by $\lambda_a^{(i-1)}$, the a priori LLR of the $(i-1)$ th iteration. The post-cursor (or past) symbols $\bar{\mathbf{s}}$ are calculated by combining the extrinsic LLR and the a priori LLR of the current iteration as $\lambda_s^{(i)} = \lambda_e^{(i)} + \lambda_a^{(i)}$, where $\lambda_a^{(i)}$ is the interleaved decoder LLR of the current iteration. The output of the HSIC unit is filtered through the equalizer to yield the (n, k) th soft symbol $\hat{s}_{n,k}$.

Using the equalized symbol to compute the extrinsic LLR requires knowledge of the distribution of $\hat{s}_{n,k}$ which is often unknown. The common approach is to simply assume that the fading channel cascaded with the equalizer is equivalent to a Gaussian noise channel between the transmit symbol and the equalized symbol, such that

$$\hat{s}_{n,k} = \mu_{n,k} s_{n,k} + v_{n,k}, \quad (2)$$

with $v_{n,k}$ being a zero-mean Gaussian noise. Conditioned on a given transmitted symbol in the modulation scheme, the equalized symbol $\hat{s}_{n,k}$ follows a Gaussian distribution with mean and variance computed from the equalizer vector and the corresponding channel vector.

The optimal equalizer coefficients are computed through the covariance matrices of the received signal vector $\mathbf{y}_k$ and the a priori soft symbols $\bar{\mathbf{s}}$, as well as the MIMO fading channel [4]. Theoretically, the optimal equalizer vector can adapt for each n and k . However, due to the high computational complexity of matrix inversion, the low-complexity approach is to design a time-invariant equalizer vector for each transmit branch n and adapt the equalizer after each Turbo iteration.

The turbo LE in [4] also incorporates a reliability ordering scheme to enhance the performance. The reliability of a soft decision is measured as the inverse of the variance of the soft decision. The idea of reliability ordering extends the layering approach in the V-BLAST to order the detection of symbols in both space and time. The V-BLAST selects the transmit branch with the strongest receiving power to detect first, while the turbo LE selects the symbol with the highest reliability to detect first. The interference of the detected symbol to other symbols in the block is then removed through the HSIC. The turbo LE with HSIC also enjoys low computational complexity and stable performance besides high data efficiency.

TD TURBO BDFE

A reliability-based block DFE (BDFE) is illustrated in Fig. 3b, where the IBI removal, channel estimation, and SIC units are similar to those used in the TD turbo LE. However, the BDFE uses two filter matrices — an FF matrix and an FB matrix — to output a block of equalized symbols $\mathbf{r}$ simultaneously for all N branches. The FF filter $\mathbf{F}$ in the BDFE has coefficients for all NL_f rows and ML_f columns, rather than the one-column FF filter vector $\mathbf{f}_n$, which is designed separately for each branch in the turbo LE scheme.

Meanwhile, the FB filter $\mathbf{B}$ is an NL_f by NL_f matrix that reconstructs the residual ISI vector $\mathbf{x}$ from the *a posteriori* soft symbols $\hat{\mathbf{s}}$ and the masked *a priori* soft symbols $\tilde{\mathbf{s}}_{n,k}$. The overall equalized symbol vector is then $\hat{\mathbf{s}} = \mathbf{r} - \mathbf{x} + \tilde{\mathbf{s}}$.

Taking advantage of the block of symbols available at the equalizer output, the BDFE utilizes the sequence-based LLR to calculate the *a posteriori* LLR $\lambda_T^{(i)}$ of each symbol and to detect the *a posteriori* soft symbols $\hat{\mathbf{s}}$. Note that we use the subscript T in the total LLR $L_T^{(i)}$ to distinguish from the sum LLR $L_s^{(i)}$ in other schemes. The post-cursor *a posteriori* symbols are also fed

BDFE suffers from huge computational complexity, especially when the block size is large, which is often the case in UWA. To reduce the complexity, two symbol-based SDFEs are developed by replacing the matrix filters with vector filters, replacing the sequence-based LLR by symbol-based LLR, and changing the FB filter structures.

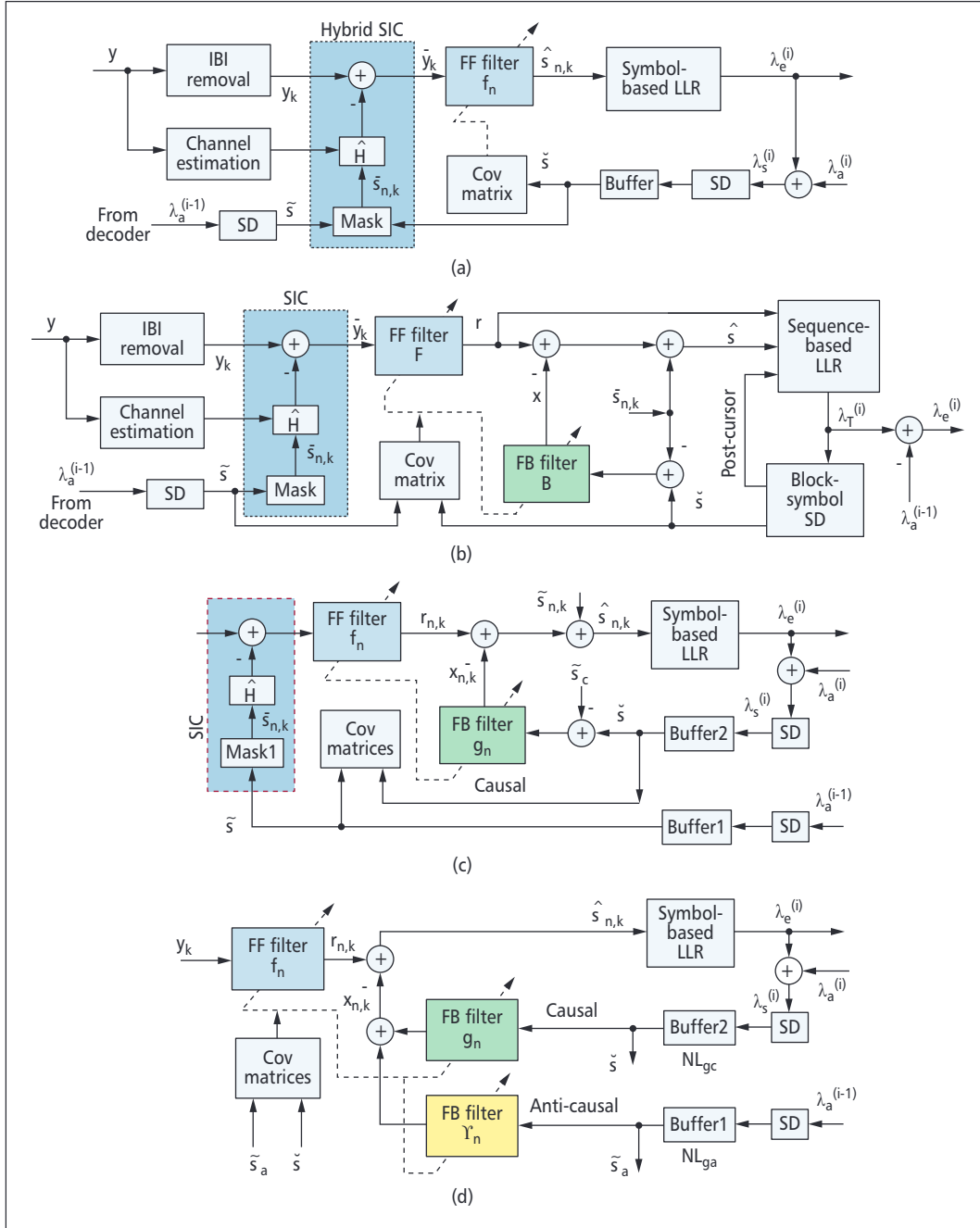


Figure 3. Time-domain turbo equalizers with the mask block set the (n, k) th element to zero. Note that turbo LE, SDFE, and SICE have N sets of FF and FB filters $\mathbf{f}_n$, $\mathbf{g}_n$, and $\mathbf{y}_n$ separately designed for each branch; while the BDFE has concatenated FF and FB filters, which can suppress CCIs from other transmit branches: a) turbo LE with symbol-based LLR and HSIC [4]; b) BDFE with sequence-based LLR [5]; c) SDFE with symbol-based LLR and SIC [6]; d) SICE with symbol-based LLR [7].

Matrix inversion and Cholesky decomposition are required to compute the optimal FF and FB filter matrices, leading to high computational complexity. The complexity can be reduced by using the same FF and FB filter matrices for all symbols in one iteration, and such a simplification results in only marginal performance degradation compared to the full adaptive design.

back to the LLR unit to aid the calculation of the total LLR $\lambda_T^{(i)}$. The extrinsic LLR $\lambda_e^{(i)}$ of the i th iteration is then $\lambda_e^{(i)} = \lambda_T^{(i)} - \lambda_a^{(i-1)}$, the difference between the total LLR of the i th iteration and the a priori LLR of the $(i-1)$ th iteration.

The optimal solution of the BDFE filters is computed from the covariance matrix of the error vector $\mathbf{e} = \mathbf{r} - \mathbf{x}$ and the covariance matrix of the a priori soft symbol vector $\tilde{\mathbf{s}}$. Matrix inversion and Cholesky decomposition are required to compute the optimal FF and FB filter matrices, leading to high computational complexity. The complexity can be reduced by using the same FF and FB filter matrices for all symbols in one iteration, and such a simplification results in only marginal performance degradation compared to the full adaptive design. In addition, the calculation of the FF and FB filters are based on the common assumption of error-free decision feedback, which eliminates the need for cross-covariance between the transmitted and equalized symbol vectors. The BDFE also calculates the accurate total LLR without the approximated Gaussian assumption (Eq. 2) of the soft decisions, which is commonly used in the turbo LE case. Compared to the TD turbo LE, the BDFE significantly reduces error propagation through the inclusion of the FB filter, and reduces processing latency through the block processing approach.

TIME-DOMAIN SDFE AND SICE

Despite its excellent performance, BDFE suffers from huge computational complexity, especially when the block size is large, which is often the case in UWA. To reduce the complexity, two symbol-based SDFEs are developed by replacing the matrix filters with vector filters, replacing the sequence-based LLR by symbol-based LLR, and changing the FB filter structures.

Figure 3c shows the structure of the SDFE, where the n th FF filter $\mathbf{f}_n$ has a length NL_f with $L_f = L_a + L_c + 1$, which is similar to the turbo LE. The FB filter $\mathbf{g}_n$ of the SDFE has a length NL_{gc} with $L_{gc} = L_2 + L_c + 1$ and L_2 being the length of the causal CIR. The a priori soft symbols $\tilde{\mathbf{s}}$ are soft decisions from the a priori LLRs of the $(i-1)$ th iteration, which are input to the SIC and covariance estimator. The a priori soft symbols of the i th iteration are available for causal taps and are input to the FB filter after subtracting $\tilde{\mathbf{s}}_c$, the causal portion of $\tilde{\mathbf{s}}$. The equalized soft symbol is then

$$\hat{s}_{n,k} = \mathbf{f}_n^H(\mathbf{y}_k - \hat{\mathbf{H}}\tilde{\mathbf{s}}_{n,k}) - \mathbf{g}_n^H(\tilde{\mathbf{s}} - \tilde{\mathbf{s}}_c) + \tilde{s}_{n,k}.$$

Intuitively, the FB filter reconstructs the postcursor interference to the (n, k) th symbol, while the precursor interference is suppressed partially by the SIC and the FF filter. The optimal solutions to the FF and FB filter vectors are derived by assuming Gaussian channel between $\hat{s}_{n,k}$ and the ground-truth symbol $s_{n,k}$ as in the turbo LE case, but with non-perfect soft decisions. The low-complexity time-invariant solutions require the auto-covariance matrices of $\mathbf{s}$ and $\tilde{\mathbf{s}}$, and the cross-covariance matrix between $\mathbf{s}$ and $\tilde{\mathbf{s}}$. However, the equalizer has no access to the transmit symbol vector $\mathbf{s}$, and the statistics of the a priori symbol vector $\tilde{\mathbf{s}}$ are used instead.

Although it has fast initial convergence, the SDFE in Fig. 3c suffers from an error floor that prevents it from reaching the matched filter bound. The main reason is that the precursor interference cancellation is intertwined in the SIC and the FF filter. This motivates the SICE [7], which uses an additional transversal filter to explicitly reconstruct the precursor interference. As shown in Fig. 3d, the FB filter γ_n is a length- NL_{ga} filter for reconstructing the interference from the precursor soft symbols detected by the a priori LLR of the $(i-1)$ th iteration $\lambda^{(i-1)}$, where $L_{ga} = L_1 + L_a + 1$. The FF filter $\mathbf{f}_n$ and FB filter $\mathbf{g}_n$ are the same as those in the SDFE, but the SIC unit is removed.

Again, the filters are jointly adapted to minimize the MSE between the equalized symbol $\hat{s}_{n,k}$ and the transmitted symbol $s_{n,k}$, and the equivalent Gaussian channel assumption is used to approximate the statistics. The solution requires the auto-covariance matrices $\Phi_{\tilde{\mathbf{s}}\tilde{\mathbf{s}}}^{\sim}$ and $\Phi_{\tilde{\mathbf{s}}\tilde{\mathbf{s}}}$, and the cross-covariance matrices $\Phi_{\tilde{\mathbf{s}}\mathbf{s}}^{\sim}$ and $\Phi_{\tilde{\mathbf{s}}\mathbf{s}}$. Without the ground-truth $\mathbf{s}$, the SICE uses the turbo LE only in the first iteration, and then uses the a priori soft symbols $\tilde{\mathbf{s}}$ in place of the ground-truth symbols to compute the cross-covariance matrices $\Phi_{\tilde{\mathbf{s}}\mathbf{s}}^{\sim}$ and $\Phi_{\tilde{\mathbf{s}}\mathbf{s}}$.

EXPERIMENTAL RESULTS OF TD TURBO EQUALIZERS

Many ocean experiments were conducted with single-carrier MIMO underwater communication systems during 2007–2010, and results were reported in several publications. Here, the performance of the four TD equalizers were evaluated by the ocean experiment SPACE08, which was conducted on the coast of Martha's Vineyard, Edgartown, Massachusetts, in October 2008. The carrier frequency was $f_c = 13$ kHz, and the symbol period was $T_s = 0.1024$ ms. The transmit pulse shaping was a square-root raised cosine filter with a roll-off factor 0.2; thus, the occupied channel bandwidth was 11.7188 kHz. The transmit equipment had four transducers, and the receiver had 12 hydrophones. The transmission frame began with a linear frequency modulation (LFM) signal, followed by three packets with quadrature PSK (QPSK), 8PSK, and 16-quadrature amplitude modulation (QAM), respectively, and ended with a trailing LFM. Each packet included a maximum-length sequence (m -sequence) of length 511, and a data payload consisting of 30,000 symbols. The coding scheme was a rate-1/2 convolutional code. An analysis of the channel scattering function revealed that the maximum Doppler spread was about 5 Hz, corresponding to a channel coherence time about 100 ms or an approximate length of 1000 taps in terms of the symbol period $T_s = 0.1024$ ms. The receiver was located at 200 m or 1000 m away from the transmitter, and the recorded signals had 45 packets for 200 m and 34 packets for 1000 m. The multipath channel lengths of both ranges were estimated as $L = 100$ taps.

The bit error rate (BER) performance of the TD equalizers are compared in Fig. 4 for 8PSK and 16-QAM transmissions. The BDFE achieved the best performance, followed by the SICE.

The SDFE sometimes performed similar to the turbo LE if not worse. The EXIT chart simulation, not shown here due to page limits, indicates that the SDFE had trouble converging with the matched filter bound, while other TD turbo equalizers do.

FREQUENCY-DOMAIN TURBO EQUALIZATION

When the channel length is large, single-carrier frequency-domain equalizers (SC-FDEs) can effectively reduce computational complexity by converting the received signals into FD via fast Fourier transform (FFT). With the overhead of FFT and inverse FFT (IFFT), the complexity of FDE is lower than that of the TDE when the equalizer length is larger than 10 [8]. Since the UWA channels are often longer than 50 taps and the equalizer lengths are longer than the channels, the computational saving of FDEs is significant. Additionally, SC-FDEs can be applied to continuous transmission without ZP or CP, which is required in OFDM systems. Since ZP or CP has to be longer than the channel length in OFDM block transmission, SC-FDEs can also achieve higher data efficiency than OFDM. This section will discuss the FD turbo LE and DFE schemes suitable for MIMO UWA channels, as shown in Fig. 5, where the FFT length K is the same as the sub-block length K_{sb} in Fig. 1b. The overlap length K_{ovlp} for receiver processing is set to be the same as the channel length l , and the channel is assumed to be time-invariant for each sub-block.

Although not shown in Fig. 5, inter-block interference (IBI) removal and a CP reconstruction block are used to preprocess the received signals when the transmitted data have no ZP or CP. The IBI from the previous sub-block is removed from the current sub-block, and the CP effect is added back to the current sub-block by reconstructing the received signals due to the last K_{ovlp} transmitted symbols of the current sub-block. The CP reconstruction renders a block-circulant channel matrix in the TD, hence enabling the FD equalizers to reduce the computational complexity. It is also assumed that the FFT unit is preceded by a serial-to-parallel buffer, and the IFFT unit is followed by a parallel-to-serial buffer. The channel estimation is done in the time domain.

FREQUENCY-DOMAIN TURBO LE

Single-carrier frequency-domain turbo linear equalizers (FD-LE) have been successfully applied to UWA MIMO communications. The basic structure of an FD turbo LE is shown in Fig. 5a, where the frequency domain channel matrix $\hat{H}$ becomes block-diagonal after the FFT, and the FD equalizer is computed by k matrix inversions of $N \times N$ small matrices instead of a huge TD channel matrix of size $(NK) \times (NK)$. However, the transmitted symbols of the current sub-block are unknown in the first iteration. Therefore, the equalizer has to treat them as zeros and their contribution in the received signal

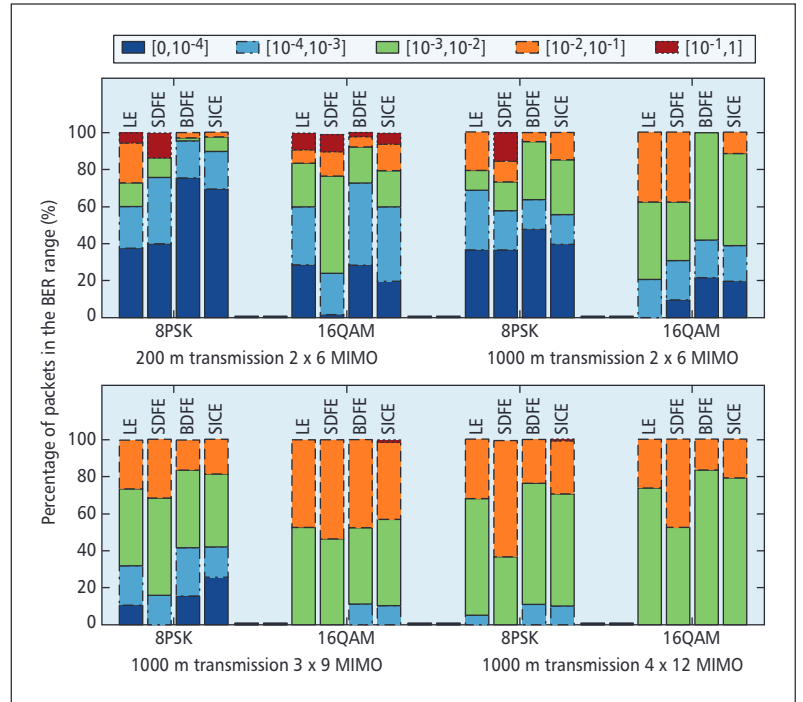


Figure 4. Error performance of the TD turbo equalizers.

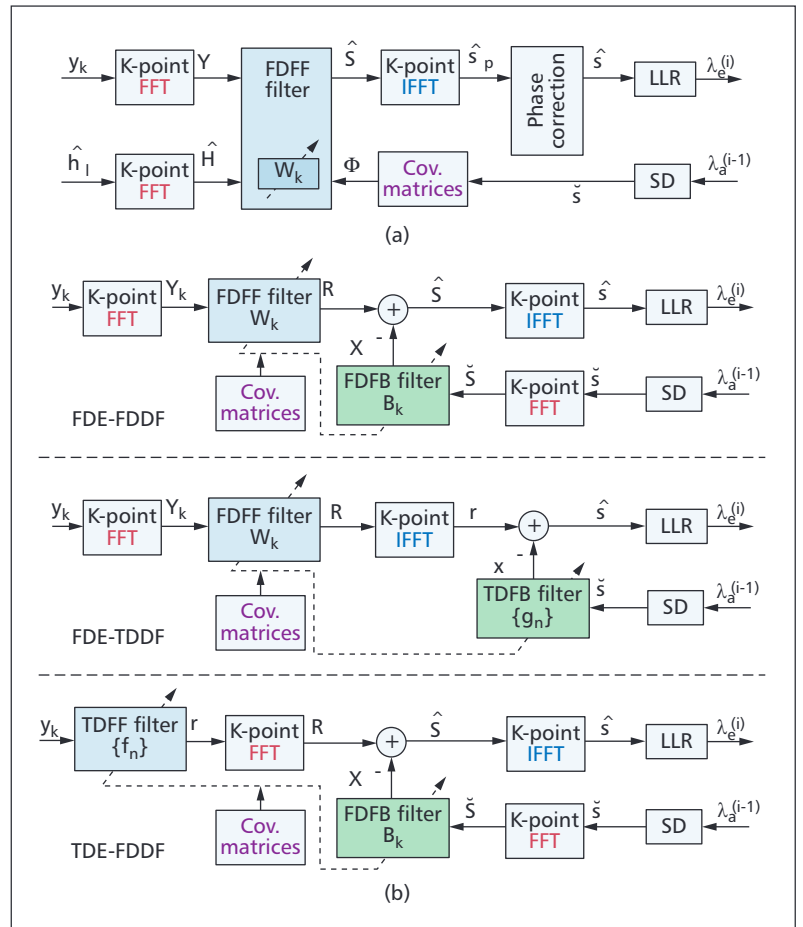


Figure 5. FD turbo equalizers, where upper case letters denote FD signals and lower case letters denote TD signals: a) FD turbo linear equalizer; b) FD turbo equalizer with decision feedback.

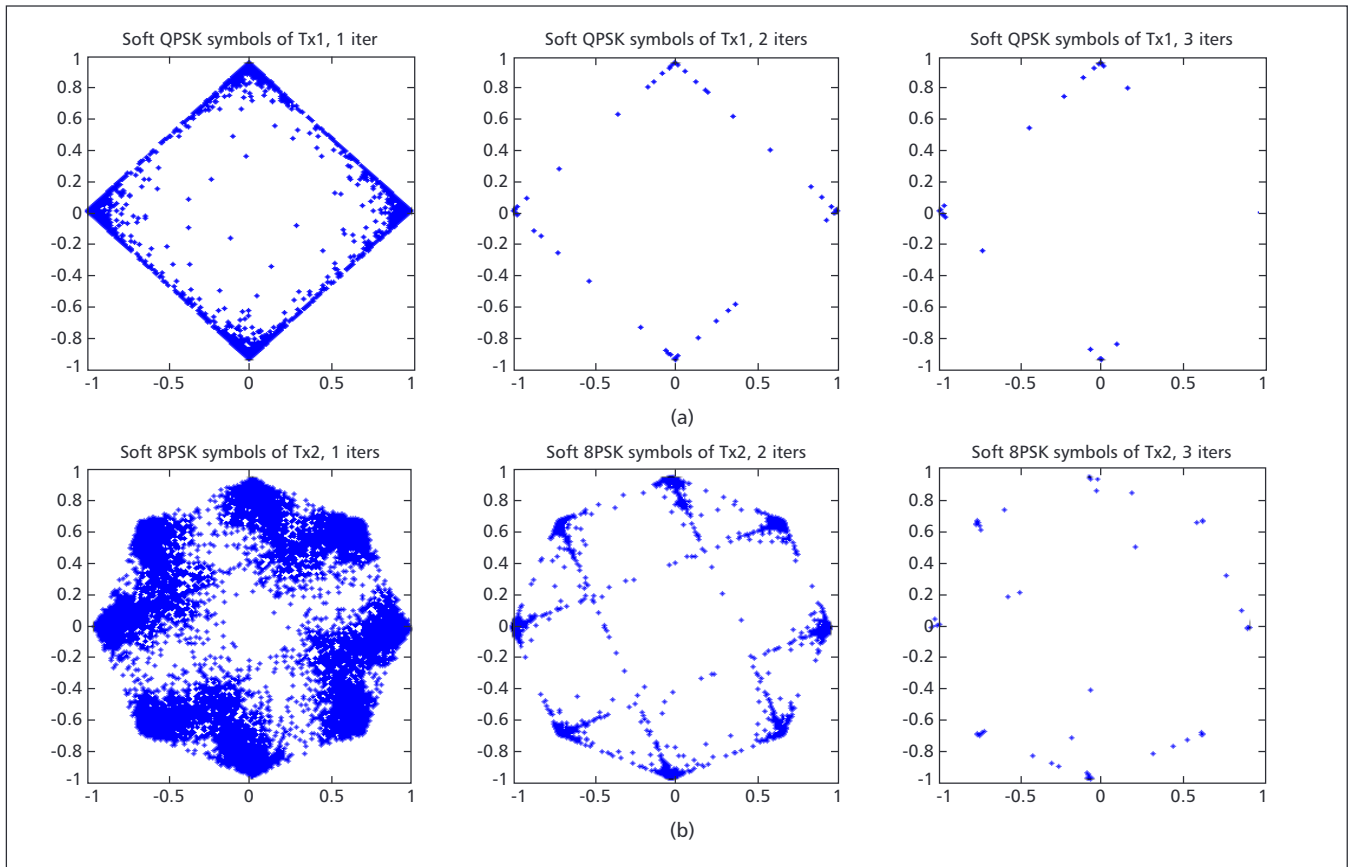


Figure 6. Scatter plots of estimated soft symbols of a packet recorded by the SPACE08 experiment with 1000 m transmission: a) QPSK; b) 8PSK.

symbols $\tilde{s}$ of the previous iteration. The FD equalizers $\mathbf{W}_k$ are designed for each frequency bin $k = 1, \dots, K$ using the FD channel matrix and the covariance matrix of the a priori soft symbols.

After the FD equalizer, the TD soft symbols $\hat{s}_p$ exhibit rapid phase rotation due to the high instantaneous Doppler spread in most UWA channels. The Doppler effect leads to non-zero entries on some of the off diagonal blocks of the FD channel matrix, and the simplified implementation of the FD equalizer results in uncompensated phase in the estimated soft symbols. A phase estimation and correction unit is added to remove the phase rotation before LLR calculation. This unit is unique for severe Doppler UWA channels, and is easy to implement by a few multiply-add operations. Other performance enhancement techniques are also available, such as layered detection with soft successive interference cancellation, time-reversal filtering, and sequence-based LLR calculation, which also increase the computational complexity.

FREQUENCY-DOMAIN TURBO DFE

The FD turbo DFEs (FD-DFEs) have two common forms: a frequency-domain equalizer with frequency-domain decision feedback (FDE-FDDF) has both FF and FB filters in the frequency domain, as shown in Fig. 5b; a frequency-domain equalizer with time-domain decision feedback (FDE-TDDF) has an FF filter in FD followed by a TD feedback filter, as shown in Fig. 5b. It is interesting to note that a block-based TDE with FDDF,

shown in Fig. 5c, is also available in the literature, and the time-domain BDFE or SDFE scheme is similar to the TDE-TDDF scheme.

For low-complexity implementation, the FF and FB filters in the FDE-FDDF scheme are designed for each frequency bin, and they require inversion of small matrices. In the FDE-TDDF scheme, the $M \times N$ FF filter $\mathbf{W}_k$ is designed for each frequency bin $k = 1, \dots, K$, and the $(NL_{gc}) \times 1$ FB filter $\mathbf{g}_n$ is designed for each transmit branch $n = 1, \dots, N$. In the TDE-FDDF scheme, the FF filters are N sets of $(ML_f) \times 1$ vectors $\mathbf{f}_n$, and the FB filters are K sets of $M \times N$ matrices $\mathbf{b}_k$. The adaptation of the filters at each iteration uses the estimated FD or TD channel matrices and the covariance matrices of the soft a priori symbols from the previous iteration.

EXPERIMENTAL RESULTS FOR FD TURBO EQUALIZERS

Many data sets from ocean experiments have been processed by the FDE-LE and FDE-FDDF schemes. To illustrate the gain of iterative detection, the scatter plots of soft QPSK and 8PSK symbols after soft decision decoding at different turbo iterations are shown in Fig. 6. It is clearly seen that turbo equalization and detection gradually separate the soft symbols and push them closer to valid constellation points. The BER performance of the FD turbo LE is similar to that of the TD turbo LE [10]. It is also noted that all the FDE soft decision feedback schemes

yield similar performance as the FD turbo LE with interference cancellation.

CONCLUSION

This article reviews four TD turbo equalizer schemes and four FD turbo equalizer schemes for single-carrier MIMO UWA communications. Their performance is compared using the SPACE08 experimental data. The turbo LEs in both TD and FD achieved satisfactory performance in severe ISI channels, and their computational complexity is affordable. Low-complexity turbo SICE and FDE-TDDF are also promising for practical implementation and can achieve better performance than the turbo LE in severe non-causal channels. The time-domain BDFE scheme achieves the best performance at the cost of high computational complexity. Future work in this area includes analysis of complexity-performance trade-offs and real-time implementation.

ACKNOWLEDGMENT

This work was supported in part by the Office of Naval Research under grants N00014-07-1-0219 and N00014-10-1-0174, and by the National Science Foundation under grants ECCS-0846486, CCF-0915846, ECCS-1202075, ECCS-1231848, and ECCS-1408316. The authors would like to thank their Ph.D. students Jun Tao, Jian Zhang, Huang Lou, Amirhossein Rafati, Longbao Wang, Zengli Yang, and Weimin Duan for producing some of the results presented in the article.

REFERENCES

- [1] A. Singer, J. Nelson, and S. Kozat, "Signal Processing for Underwater Acoustic Communications," *IEEE Commun. Mag.*, vol. 47, no. 1, Jan. 2009, pp. 90–96.
- [2] C. Laot, A. Glavieux, and J. Labat, "Turbo Equalization: Adaptive Equalization and Channel Decoding Jointly Optimized," *IEEE JSAC*, vol. 19, no. 9, 2001, pp. 1744–52.
- [3] B. Li et al., "Multicarrier Communication Over Underwater Acoustic Channels with Nonuniform Doppler Shifts," *IEEE J. Ocean. Eng.*, vol. 33, no. 2, 2008, pp. 198–209.
- [4] J. Tao et al., "Enhanced MIMO LMMSE Turbo Equalization: Algorithm, Simulations, and Undersea Experimental Results," *IEEE Trans. Signal Processing*, vol. 59, no. 8, Aug. 2011, pp. 3813–23.
- [5] J. Tao et al., "Robust MIMO Underwater Acoustic Communications Using Turbo Block Decision-Feedback Equalization," *IEEE J. Ocean. Eng.*, vol. 35, no. 4, Oct. 2010, pp. 948–60.
- [6] A. Rafati, H. Lou, and C. Xiao, "Soft-Decision Feedback Turbo Equalization for LDPC-Coded MIMO Underwater Acoustic Communications," *IEEE J. Oceanic Eng.*, vol. 39, no. 1, Jan. 2014, pp. 90–99.
- [7] J. Wu, L. Wang, and C. Xiao, "Low Complexity Soft-Interference Cancellation Turbo Equalization for MIMO systems with Multilevel Modulations," *IET Commun.*, vol. 9, no. 5, 2015, pp. 728–35.
- [8] N. Benvenuto et al., "Single Carrier Modulation with Nonlinear Frequency Domain Equalization: An idea Whose Time Has Come Again," *Proc. IEEE*, vol. 98, no. 1, Jan 2010, pp. 69–96.
- [9] J. W. Choi et al., "Adaptive Linear Turbo Equalization over Doubly Selective Channels," *IEEE J. Ocean. Eng.*, vol. 36, no. 4, Oct. 2011, pp. 473–89.
- [10] J. Zhang and Y. Zheng, "Frequency-Domain Turbo Equalization with Soft Successive Interference Cancellation for Single Carrier MIMO Underwater Acoustic Communications," *IEEE Trans. Wireless Commun.*, vol. 10, no. 9, Sept. 2011, pp. 2872–82.

BIOGRAPHIES

YAHONG ROSA ZHENG [M'03, SM'07, F'15] (zhengyr@mst.edu) received her B.S. degree in electrical engineering from the University of Electronic Science and Technology of China, Chengdu, in 1987, her M.S. degree in electrical engineering from Tsinghua University, Beijing, China, in 1989, and her Ph.D. degree from Carleton University, Ottawa, Ontario, Canada, in 2002. She was an NSERC postdoctoral fellow from January 2003 to April 2005 at the University of Missouri-Columbia. In fall 2005, she joined the Department of Electrical and Computer Engineering, Missouri University of Science and Technology, Rolla, where she is currently a professor. Her research interests include array signal processing, wireless communications, and wireless sensor networks. She has served as a Technical Program Committee (TPC) member for many IEEE international conferences, including IEEE Vehicular Technology, IEEE GLOBECOM, IEEE ICC, IEEE Wireless Communications and Networking Conference, and others. She served as a Wireless Communications Symposium Co-Chair for ICC 2014 and GLOBECOM 2012. She also served as an Associate Editor of *IEEE Transactions on Wireless Communications*, 2006–2008. She is currently an Associate Editor of *IEEE Transactions on Vehicular Technology*. She was the recipient of an NSF CAREER Award in 2009.

JINGXIAN WU [S'02, M'06] (wuj@uark.edu) received his B.S. degree from Beijing University of Aeronautics and Astronautics, China, in 1998, his M.S. degree from Tsinghua University, Beijing, in 2001, and his Ph.D. degree from the University of Missouri at Columbia in 2005, all in electrical engineering. He is currently an associate professor with the Department of Electrical Engineering, University of Arkansas, Fayetteville. His research interests mainly focus on signal processing for large-scale networks and wireless communications, including energy-efficient information sensing and processing, green communications, and statistical data analytics. He served as Co-Chair of the 2012 Wireless Communication Symposium of IEEE ICC, and the 2009 and 2015 Wireless Communication Symposia of IEEE GLOBECOM. He served as an Associate Editor of *IEEE Transactions on Vehicular Technology* from 2007 to 2011. He is also an Editor of *IEEE Transactions on Wireless Communications* and an Associate Editor of *IEEE Access*.

CHENGSHAN XIAO [M'99, SM'02, F'10] (xiaoc@mst.edu) earned a B.S. degree in electronic engineering from the University of Electronic Science and Technology of China in 1987, an M.S. degree in electronic engineering from Tsinghua University in 1989, and a Ph.D. in electrical engineering from the University of Sydney in 1997. Since 2010, he has been a professor with the Department of Electrical and Computer Engineering at Missouri University of Science and Technology. He is currently serving as a program director at USA National Science Foundation through inter-governmental personnel assignment. Previously, he was a senior member of scientific staff with Nortel Networks, Ottawa, Canada, a faculty member at Tsinghua University, the University of Alberta, Edmonton, Canada, and the University of Missouri — Columbia. He also held visiting professor positions in Germany and Hong Kong. His research interests include wireless communications, signal processing, and underwater acoustic communications. He is the holder of three U.S. patents. He is the Director of Conference Publications of IEEE ComSoc and a member of the IEEE ComSoc Fellow Evaluation Committee. Previously, he served as an elected member of the IEEE ComSoc Board of Governors, and a Distinguished Lecturer of both IEEE ComSoc and the IEEE Vehicular Technology Society. He has also served as an Editor, Area Editor, and Editor-in-Chief of *IEEE Transactions on Wireless Communications*, and an Associate Editor of *IEEE Transactions on Vehicular Technology* and *IEEE Transactions on Circuits and Systems-I*. He was Technical Program Chair of IEEE ICC 2010, Cape Town, South Africa. He served as the founding Chair of the IEEE Technical Committee on Wireless Communications. He is a recipient of the 2014 Humboldt Research Award from the Alexander von Humboldt Foundation, and the 2014 Joseph LoCicero Award from IEEE ComSoc.

It is clearly seen that turbo equalization and detection gradually separated the soft symbols and pushed them closer to valid constellation points. The BER performance of the FD turbo LE is similar to the TD Turbo LE. It is also noted that all the FDE soft decision feedback schemes yield similar performance as the FD turbo LE with interference cancellation.